

Find $\sup(A)$, $\max(A)$, $\inf(A)$, $\min(A)$

① $m \neq 0$

a) $A = \left\{ \frac{1}{n}, n \in \mathbb{N}^* \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots \right\}$

$$ub(A) = [1, \infty) \Rightarrow \sup(A) = 1 \in A \Rightarrow \exists \max A = 1.$$

$$lb(A) = [-\infty, 0] \Rightarrow \inf(A) = 0 \notin A \Rightarrow \nexists \min A.$$

? $1 \in \text{lb}(A), 1 \leq a, (\forall) a \in A$ False.

? Let $x > 0$. Then $\exists m \in \mathbb{N}$ s.t. $x > \frac{1}{m}$, $m > \frac{1}{x}$, $m = \left\lceil \frac{1}{x} \right\rceil + 1$.

$$\boxed{\text{mim}(A) = \text{lb}(A) \cap A} = \emptyset \text{ (here)}$$

b) $A = \{x \in \mathbb{Q} \mid x^2 \leq 2\} = [-\sqrt{2}, \sqrt{2}] \cap \mathbb{Q}$

$$ub(A) = [\sqrt{2}, \infty)$$

$$\Delta_b(A) = (-\infty, -\sqrt{2}]$$

$$\sup(A) = \sqrt{2} \notin A \Rightarrow \nexists \max$$

$$\inf f(A) = -\sqrt{2} \notin A \Rightarrow \nexists \min$$

! Every set that is bounded above has a sup.

an inf.

THEOREM.

$A \subseteq \mathbb{R}$ bounded set.

$$(1) (\dagger) \varepsilon > 0, \exists x \in A \text{ s.t. } \sup A - \varepsilon < x \quad (P_1)$$

(2) (4) $\varepsilon > 0, \exists x \in A$ s.t. $x < \inf(A) + \varepsilon$ (P2)

Proving (1) :

Let $\varepsilon > 0$. Assume that (*) $x \in A$, $\sup A - \varepsilon \geq x$ (!P₁)

$$\Rightarrow \sup A - \varepsilon \in \text{ub } A$$

and

$$\sup A - \varepsilon < \sup A \quad (*)$$

But $\sup A$ is the smallest ub:

$$(\forall) u \in \text{ub}(A), u \geq \sup(A) \quad (**)$$

$(*)$, $(**)$ $\Rightarrow$ CONTRADICTION. $\Rightarrow P_1$ is true.

Proving (2):

Let $\varepsilon > 0$. Assume that $(\forall) x \in A : x \geq \inf(A) + \varepsilon$ ($\neg P_2$)

$$\Rightarrow \inf(A) + \varepsilon \in \text{lb}(A)$$

and

$$\inf A + \varepsilon > \inf (*)$$

But $\inf A$ is the largest lb:

$$(\forall) u \in \text{lb}(A), u \leq \inf A \quad (**)$$

$(*)$, $(**)$ $\Rightarrow$ CONTRADICTION $\Rightarrow P_2$ is true.

! Bounded set $\rightarrow$ always $\inf \leq \sup$
with equality when we have only one element in the set.

$$(\text{Ex: } A = \{5\} \Rightarrow \inf A = \sup A = 5)$$

PROPOSITION

$A \subseteq B \subseteq \mathbb{R}$ nonempty bounded sets.

$$\inf B \leq \inf A \leq \sup A \leq \sup B$$

$$\begin{array}{lcl} \text{Ex: } A = (2, 3) & \begin{cases} \inf A = 2 \\ \sup A = 3 \end{cases} & \\ B = [1, 5) & \begin{cases} \inf B = 1 \\ \sup B = 5 \end{cases} & 1 \leq 2 \leq 3 \leq 5 \end{array}$$

$$\sup(A \cup B) = \max \{ \sup A, \sup B \}$$

$$\inf(A \cup B) = \min \{ \inf A, \inf B \}$$

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$$

↳ If A is not bounded above: $\sup A = \infty$

↳ If A is not bounded below: $\inf A = -\infty$

! $\emptyset$ is bounded by any number.

$$\sup(\emptyset) = -\infty$$

$$\inf(\emptyset) = \infty$$

Neighbourhoods:

$$V \subseteq \mathbb{R} \text{ is a neighbourhood of } x \in \mathbb{R} \text{ if } \exists \varepsilon > 0 \text{ s.t. } (x - \varepsilon, x + \varepsilon) \subseteq V$$

• neighbourhood of ∞ : $\exists a \in \mathbb{R} \text{ s.t. } (a, \infty) \subseteq V$.

• neighbourhood of $-\infty$: $\exists a \in \mathbb{R} \text{ s.t. } (-\infty, a) \subseteq V$.

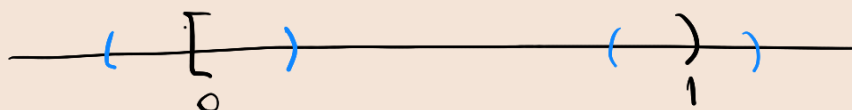
$$\mathcal{N}(x) = \{ V \subseteq \mathbb{R} \mid V \text{ neighbourhood of } x \}$$

Interior:

$$\text{int}(A) = \{ x \in \mathbb{R} \mid \exists V \in \mathcal{N}(x) \text{ s.t. } V \subseteq A \}$$

$$\text{Ex: } A = [0, 1)$$

$$\text{int}(A) = (0, 1)$$



Closure:

$$\mathcal{C}(A) = \{x \in \mathbb{R} \mid \exists V \in \mathcal{V}(x) \text{ s.t. } \underline{V \cap A \neq \emptyset}\}$$

Ex: $[0, 1) = A$

$\mathcal{C}(A) = [0, 1]$.

